

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Monocarboxylic Acid Transport (GO:001571)	0.25136227	50	6.202e-10	3.331e-06	SLC5A12:15 SLC16A5:222 SLC10A3:282 SLC5A8:439 SLC01A2:474 SLC16A9:568
Organic Hydroxy Compound Transport (GO:0)	0.21918957	36	5.435e-06	1.460e-02	SLC10A3:282 SLC5A8:439 SLC01A2:474 SLC22A1:690 ABCG5:834 SLC01B1:860
Lysosome Localization (GO:003418)	-0.23448608	27	2.500e-05	4.76e-02	SPAG9:18 GRP:659 NDEL:1.743 KIT:807 HP56:875 BORCS7:1032
Bile Acid And Bile Salt Transport (GO:00)	0.28682788	16	7.157e-05	9.611e-02	SLC10A3:282 SLC01A2:474 SLC01B1:860 SLC51B:931 SLC51A:939 SLC10A6:1062
Organic Cation Transport (GO:0015695)	-0.0083671	30	1.420e-04	1.525e-01	SLC22A16:24 SLC44A5:401 SLC6A12:596 SLC22A1:690 SLC18B1:994 RALBP1:1037
Positive Regulation Of Cell Population P	-0.05836496	328	3.177e-04	2.844e-01	MAPK15:35 GDNF:36 KMT2D:75 ILSRA:88 ZFPM2:99 IGF1R:101
tRNA Methylation (GO:0030488)	-0.17542527	34	4.045e-04	3.103e-01	TARBP1:259 TRMT10A:592 TRMT9B:1242 LCMT2:1292 METTL28:1489 TRMT10C:1800
Carboxylic Acid Transport (GO:0046942)	0.14291862	47	7.095e-04	4.234e-01	SLC5A12:15 SLC16A5:222 SLC36A2:227 SLC6A6:519 SLC36A1:564 SLC16A9:568
Small Nucleolar Ribonucleoprotein Comple	-0.32775028	9	6.632e-04	4.234e-01	TNIP1:139 RUVBL2:704 SHQ1:1577 RUVBL1:1640 PH1D1:1854 ZNHIT3:2199
RNA Methylation (GO:0001510)	0.14016765	48	7.917e-04	4.252e-01	TARBP1:259 TRMT10A:592 FTSJ3:618 TRMT9B:1242 LCMT2:1292 CMTFR1:1326
Energy Derivation By Oxidation Of Organi	-0.20015078	22	2.161e-03	4.372e-01	COX15:42 COX6A1:345 PPARGCA1A:1141 COX4I1:1162 CYC1:1463 NDUF5A:1526
Mast Cell Activation Involved In Immune	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Mast Cell Degranulation (GO:0043303)	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Mast Cell Mediated Immunity (GO:002448)	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551
Monoatomic Cation Homeostasis (GO:005508)	0.11946207	61	1.273e-03	4.372e-01	CYBRD1:66 KCNH2:133 SLC12A8:160 CLDN16:186 LETM1:383 CNNM4:537
Neuron Differentiation (GO:0030182)	-0.08695278	114	1.387e-03	4.372e-01	RTN4:132 USH2A:265 SPINK5:358 TENM1:431 ERBB4:453 WNT7B:470
Peptidyl-Tyrosine Phosphorylation (GO:00)	-0.12634054	57	9.858e-04	4.372e-01	IGF1R:101 ERBB4:453 AB13:473 EFEMP1:593 DDR1:620 RELN:634
Ubiquitin-Dependent Protein Catabolic Pro	-0.06180434	226	1.465e-03	4.372e-01	AMBRA1:34 TBL1XR1:40 PSMA7:138 USP33:180 PSMD7:184 PSB1:191
Positive Regulation Of T Cell Activation	-0.10627397	74	1.606e-03	4.539e-01	CD276:373 THY1:500 TNFSF11:669 CD274:721 ZP4:784 IL21:845
Stem Cell Differentiation (GO:0048863)	-0.17775037	26	2.176e-03	4.609e-01	LMBR1L:55 TP53:297 FOXO4:499 KIT:807 FGFR2:955 MEOX1:1014
Calcium-Independent Cell-Cell Adhesion V	0.26764411	11	2.118e-03	4.783e-01	CLDN16:186 CLDN8:211 CLDN18:460 CLDN20:1460 CLDN7:1466 CLDN23:2614
Intracellular Monoatomic Ion Homeostasis	0.20519185	19	1.967e-03	4.783e-01	CLDN16:186 LETM1:383 CNNM4:537 SLC4A1:670 RHAG:914 RHCG:1513
Lipid Transport (GO:0006869)	0.10030826	78	2.239e-03	4.783e-01	EHADH:116 ABCB11:144 ABCA2:161 SLC10A3:282 SLC10A3:1369 PRKDC:1509 SHLD2:1655
Positive Regulation Of Double-Strand Bre	0.28281818	10	1.959e-03	4.783e-01	KMT5C:924 WRAP53:967 SMCHD1:1003 KMT5B:1369 PRKDC:1509 SHLD2:1655
Regulation Of Synaptic Vesicle Exocytosis	-0.25404122	12	2.315e-03	4.783e-01	RAP1A:1 VPS18:77 RIMS1:481 SYN1:586 RIMS2:936 SINAPIN:1457
Ribosome Biogenesis (GO:0042654)	-0.08360945	112	2.306e-03	4.783e-01	TNIP1:139 PAK1IP1:195 NOB1:372 C1NP:443 WDR75:463 BYSL:477
Positive Regulation Of Defense Response	0.0390261	7	2.433e-03	4.836e-01	PGC:5 EMILIN2:215 GRN:723 KLU5:1274 KLU7:1275 EMILIN1:5581
Positive Regulation Of Transcription By	-0.03595721	628	2.521e-03	4.836e-01	ESRRG:4 NR5A2:122 GDNF:36 ABHD148:38 TBL1XR1:40 STING1:41
rRNA Metabolic Process (GO:0016072)	-0.11060408	62	2.637e-03	4.884e-01	NOB1:372 BYSL:477 NOP58:748 DD54:984 XRN2:1125 ESF1:1134
Antigen Receptor-Mediated Signaling Path	-0.09032681	90	3.228e-03	4.917e-01	CD276:373 NFATC2:457 TRAF6:497 THY1:500 STOML2:581 EIF2B2:625
Cardiac Right Ventricle Morphogenesis (G	-0.42305169	4	3.387e-03	4.917e-01	ZFPM2:99 GATA4:909 JAG1:960 SOX4:1926 NA NA
Centromeric Sister Chromatid Cohesion (G	0.42507066	4	3.238e-03	4.917e-01	MEIKIN:319 SGO1:740 BUB1:1114 BUB1B:1623 NA NA
Ear Morphogenesis (GO:004271)	-0.22680159	14	3.309e-03	4.917e-01	TIFAB:68 USH1C:7 FGR2:955 OSR2:2716 EPHB2:2746 NOG:2781
Fat-Soluble Vitamin Metabolic Process (G	0.21310895	16	3.174e-03	4.917e-01	CYP11A1:82 CYP2R1:92 CYP1A1:318 LRA1:678 GGCX:1502 KDM5A:2026
Macromolecule Biosynthetic Process (GO:0	-0.08395076	104	3.178e-03	4.917e-01	TNIP1:139 RPL28:224 RPLP0:234 MRPL27:352 SARS1:450 RP55:541
Regulation Of JNK Cascade (GO:0046328)	-0.08920613	93	3.018e-03	4.917e-01	MAP3K11:80 IGF1R:101 CYLD:133 WNT7B:470 TRAF6:497 HMGB1:660
Regulation Of Type B Pancreatic Cell Pro	-0.38375568	5	2.962e-03	4.917e-01	SGPP2:677 PTPRN1:2221 NR4A1:1516 NR4A3:1537 NUPR1:2446 NA
Negative Regulation Of SMAD Protein Sign	0.37634528	5	3.565e-03	5.039e-01	SLC2A10:50 PBLD:158 VEPH1:894 LRPI:1341 CILP:5418 NA
Acid Secretion (GO:0046717)	0.31623173	7	3.766e-03	5.057e-01	SLC22A16:24 DRD3:837 SLC51B:931 SLC51A:939 GHRL:1547 SLC26A6:4198
Positive Regulation Of Isozyme Switching	0.23272398	13	3.676e-03	5.057e-01	HMCES:63 TGFBI:269 PAXIP1:902 KMT5C:924 KMT5B:1369 SHLD2:1655

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BLAOCK_ALZHEIMERS_DISEASE_DN	-0.06204481	766	8.308e-09	5.380e-05	PTGES3:10 CDK17:15 UBL3:23 PLK2:54 CSNK1G3:60 ATP5F1:81
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	-0.16876670	80	1.868e-07	6.050e-04	PSMA7:138 RNPS1:142 USP33:180 PSMD7:184 RPL28:224 RPLP0:234
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	-0.21293218	46	5.935e-07	1.281e-03	PSMA7:138 PSMD7:184 TP53:297 UBB:399 SEM1:702 PSMF1:809
HOUNKPE_HOUSEKEEPING_GENES	-0.05424055	712	1.129e-06	1.828e-03	SPAG9:18 TBL1XR1:40 COX15:42 BAP1:45 AP3M1:53 TMEM2:1959
REACTOME_SIGNALING_BY_WNT	-0.11009639	162	1.419e-06	1.838e-03	PLCB3:51 KMT2D:75 PSMA7:138 VPS35:144 PRICKLE1:163 PRKG1:167
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	-0.12834661	113	2.545e-06	2.354e-03	PSMA7:138 RNPS1:142 USP33:180 PSMD7:184 PFN1:202 RPL28:224
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.12112995	128	2.328e-06	2.354e-03	PHC1:73 KMT2D:75 PSMA7:138 PSMD7:184 CBX6:223 UBB:399
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	-0.07854398	297	3.613e-06	2.925e-03	RAP1A:1 POLR2E:21 TBL1XR1:40 MAP3K11:80 PSMA7:138 PSMD7:184
REACTOME_DEGRADATION_OF_GLI1_BY_THE_PROT	-0.23125302	32	6.023e-06	4.334e-03	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 NUMB:738 PSMF1:809
REACTOME_REGULATION_OF_RNA_STABILITY_BY	-0.18948137	46	8.856e-06	5.735e-03	PSMA7:138 PSMD7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809
MILL_PSEUDOPODIA_HAPTOTAXIS_UP	-0.0777197	328	1.182e-05	6.958e-03	RAP1A:1 AP3M1:53 CSNK1G3:60 PLEKHA8:84 ZRANB2:87 EIF2A:116
REACTOME_DEVELOPMENTAL_BIOLOGY	-0.04894056	681	1.700e-05	9.172e-03	NR5A2:12 SPAG9:18 POLR2E:21 EFNA5:29 TRPC5:31 GDNF:36
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	-0.18262151	46	1.848e-05	9.206e-03	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_NEGATIVE_REGULATION_OF_NOTCH4_S	0.23034815	27	3.449e-05	1.507e-02	PSMA7:138 PSMD7:184 UBB:399 TACC3:605 SEM1:702 PSMF1:809
CAIRO_HEPATOBLASTOMA_DN	0.08602525	196	3.491e-05	1.507e-02	CPT2:59 TMEM131L:77 SLC1A1:84 ABCB11:144 ACA2:239 ACADM:307
REACTOME_TCF_DEPENDENT_SIGNALING_IN_RCP	-0.11105458	115	4.022e-05	1.557e-02	KMT2D:75 PSMA7:138 PSMD7:184 KREMN2:381 UBB:399 SEM1:702
REACTOME_ASYMMETRIC_LOCALIZATION_OF_PSC	-0.19712965	36	4.292e-05	1.557e-02	PSMA7:138 PRICKLE1:163 PSMD7:184 UBB:399 PARDBA:642 SEM1:702
REACTOME_DEFECTIVE_CFTR_CAUSES_CYSTIC_FI	-0.20577752	33	4.326e-05	1.557e-02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	-0.07968579	219	5.150e-05	1.755e-02	RAP1A:1 GDNF:36 MAP3K11:80 ILSRA:88 PSMA7:138 PSMD7:184
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.14034637	69	5.647e-05	1.829e-02	ITGA5:37 SMADE6:130 PSMA7:138 PSMD7:184 UBB:399 SEM1:702
REACTOME_REGULATION_OF_RAS_BY_GAPS	-0.18789741	38	6.173e-05	1.904e-02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_AUFI_HNRNP_D0_BINDS_AND_DESTABI	-0.20721344	31	6.567e-05	1.933e-02	PSMA7:138 PSMD7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.07801675	219	7.373e-05	2.076e-02	SLC5A12:15 SLC22A16:24 SLC2A10:50 SLC1A1:84 RSC1A1:142 SLC36A2:227
HSIAO_HOUSEKEEPING_GENES	-0.08033602	204	8.084e-05	2.181e-02	ABLIM1:62 CD34:64 AARS1:170 PSMD7:184 PFN1:202 RPL28:224
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	-0.13658717	69	8.884e-05	2.301e-02	PSMA7:138 PSMD7:184 UBB:399 NFATC2:457 SEM1:702 MALT1:728
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	-0.05273123	470	1.026e-04	2.447e-02	RAP1A:1 ASB15:13 CD34:64 CD4:70 FBXW9:137 PSMA7:138
WIP_NRF2_PATHWAY	0.11711941	92	1.059e-04	2.447e-02	SLC5A12:15 SLC2A10:50 TXNRD1:237 SLC2A11:264 TGFBI:269 SLC2A7:315
KEGG_LYSINE_DEGRADATION	0.18256884	38	9.922e-05	2.447e-02	NSD2:72 EHADH:116 EHM1:283 NSD3:352 GCDH:418 SETD1B:859
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04803619	566	1.100e-04	2.456e-02	TRPM6:6 TRPM5:12 SLC5A12:15 SLC22A16:24 ADCY1:27 ATP9B:38
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.12714593	77	1.168e-04	2.521e-02	SLC22A16:24 RSC1A1:142 SLC14A2:332 SLC13A2:389 SLC4A45:401 SLC6A6:519
REACTOME_STABILIZATION_OF_P53	-0.18888309	34	1.390e-04	2.905e-02	PSMA7:138 PSMD7:184 TP53:297 MDM2:317 UBB:399 SEM1:702
KIM_ALL_DISORDERS_CALB1_CORR_UP	-0.06009307	337	1.638e-04	3.315e-02	POLR2E:21 UBL3:23 BAP1:45 PLK2:54 RTN4:132 ATP6V1D:152
REACTOME_SIGNALING_BY_NOTCH4	-0.15317167	49	2.099e-04	4.120e-02	PSMA7:138 PSMD7:184 UBB:399 TACC3:605 MAML3:614 SEM1:702
REACTOME_NEDDYATION	-0.08882418	146	2.191e-04	4.174e-02	ASB15:13 FBXW9:137 PSMA7:138 PSMD7:184 UBB:399 PSB1:191 COP56:346
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF	-0.10083603	45	2.285e-04	4.228e-02	PSMA7:138 PSMD7:184 UBB:399 NFATC2:457 SEM1:702 MALT1:728
REACTOME_UP_SPECIFIC_PROCESSING_PROTEASE	-0.15831610	111	2.677e-04	4.686e-02	USP19:103 CYLD:133 PSMA7:138 USP33:180 PSMD7:184 WDR20:240
REACTOME_REGULATION_OF_RUNX2_EXPRESSION	-0.16868661	39	2.650e-04	4.686e-02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_PCP_CE_PATHWAY	-0.14570295	52	2.814e-04	4.795e-02	PSMA7:138 PRICKLE1:163 PSMD7:184 PFN1:202 CLTC:348 UBB:399
ACEVEDO_LIVER_TUMOR_VS_NORMAL_ADJACENT_T	-0.04447811	579	2.947e-04	4.893e-02	AIF1:90 PKP2:97 PPP1CA:104 TBL2:109 EIF2A:116 PRRC2:117
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	-0.18959056	30	3.272e-04	5.105e-02	PSMA7:138 PSMD7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cholestasis	0.08695886	229	6.578e-06	6.426e-02	AFP:14 CLU:42 SLC23A1:76 ABCB11:144 KANK1:213 HSD17B4:385
Metabolic Bone Disorder	0.20114153	29	1.792e-04	5.833e-01	CYP2R1:92 ANKRD11:136 MMP2:514 PTH:520 CA2:701 GLB1:833
Nephrocalcinosis	0.16215230	45	1.702e-04	5.833e-01	TRPM6:6 CYP2R1:92 MPP2:514 CLDN16:186 ATP6V0A4:312 CDC73:470
Anemia, hereditary spherocytic hemolytic	0.04105786	5	1.476e-03	6.763e-01	SPTA1:123 ANK1:266 SLCRA1:670 EPB42:1416 SPTB:3113 NA
Cholestasis, chronic	0.36238981	5	5.012e-03	6.763e-01	ABCB11:144 AKR1D1:628 ABCB4:716 ATP8B1:1609 F2R:5523 NA
Cockayne Syndrome, Type II	-0.20418954	15	6.194e-03	6.763e-01	TP53:297 ERCC48:85 RIMS2:936 OGG1:938 IGF1:979 NLFEE:1130
Inclusion Body Myopathy, Sporadic	-0.11246798	47	7.706e-03	6.763e-01	KCNH8:166 KCNH4:215 NT5C1A:295 UBB:399 TIMP1:464 OPTN:900
Lipomatosis, Multiple	-0.17794074	24	2.575e-03	6.763e-01	GDNF:36 BAP1:45 CD34:64 STAT5A:284 TP53:297 MDM2:317
Liposarcoma, Dedifferentiated	0.06372566	27	2.350e-03	6.763e-01	CD34:64 IGF1R:101 TP53:297 MDM2:317 HOKA5:355 RB1:885
Polyarticular Juvenile Idiopathic Arthri	-0.17879728	17	7.333e-03	6.763e-01	LNPEP:1401 IL2RA:1451 LTB1R:1630 CCR3:1790 STAT4:1869 STAT1:2047
Psychoses, Drug	0.23994118	15	1.297e-03	6.763e-01	CAMK2B:106 RGS9:183 RGS9B:578 CALR:966 DTPB:1862
Thymoma, type C	-0.20718557	14	7.286e-03	6.763e-01	IGF1R:101 TP53:297 HMGB1:660 KIT:807 ERBB2:1571 CD70:1763
22q11 Deletion Syndrome	0.22230733	16	2.085e-03	6.763e-01	HIRA:13 ZDHHC8:48 COMT:391 ARVCF:805 CHRD:948 FGF8:2009
Acute kidney injury	0.10032471	62	6.371e-03	6.763e-01	CLU:42 ALB:79 TGFBI:269 A2M:454 DKKE:498 AMN:522
Adenocarcinoma of rectum	0.15880843	24	7.107e-03	6.763e-01	ENTPD1:46 KRT20:486 NTSE:871 MYD88:1141 FOXO3:1202 CDC15:10
Anxiety state	0.24859235	11	4.311e-03	6.763e-01	IL1R2:98 COMT:391 TNF:1025 IL1A:1802 BDNF:2052 GDE1:2140
Atherosclerosis	0.13230981	39	4.281e-03	6.763e-01	TRPV1:295 PNIP:561 HP:777 ABCG5:834 CNR2:945 TNF:1025
Benign melanocytic nevus	-0.11695944	48	5.246e-03	6.763e-01	BAP1:45 TP53:297 MDM2:317 CDK6:468 KIT:807 TUFM:869
Benign Neoplasm	0.1236498	47	4.390e-03	6.763e-01	KCNH2:133 BRCA1:243 TRPV1:295 GPR55:567 ATM:582 VDR:1208
Benign recurrent intrahepatic cholestasi	0.47946095	6	3.162e-03	6.763e-01	ABCB11:144 GGTL3C:406 ATP8B1:1609 GGT2:2467 GGT1:3285 GGTCL1:3482
Bilirubin measurement	0.22390196	15	2.686e-03	6.763e-01	PLA2R1:290 SLC01A2:474 DGKJ:719 MROH4:734 SLC01B1:860 DCHS2:1352
Broad chest	-0.32203611	6	6.302e-03	6.763e-01	XYLT2:342 CHST3:1288 MAP2K1:1722 TRAPP2C:18